***Brock Biology of Microorganisms, 15e* (Madigan et al.)**
Chapter 12 Biotechnology and Synthetic Biology

12.1 Multiple Choice Questions

1) If a foreign gene is cloned into an expression host, it is important that the host itself
A) not produce the protein being studied.
B) produce the protein in larger amounts than the vector.
C) repress the genetic expression being studied.
D) produce signal proteins to tag the host protein.
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.3

2) Cells that have "insertional inactivation" of the lacZ gene are
A) blue.
B) white.
C) yellow.
D) fluorescent green.
Answer: B
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.2

3) Cosmids are a type of
A) bacterial artificial chromosomes (BACs).
B) cloning vector.
C) heat stable polymerase.
D) RNA/DNA hybrid.
Answer: B
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.2

4) Expression vectors are designed to ensure that ________ can be efficiently ________.
A) mRNA / transcribed
B) DNA / transcribed
C) mRNA / translated
D) DNA / translated
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.3

5) A(n) ________ gene is a gene that encodes a protein that is easy to detect and assay.
A) encoder
B) translational
C) reporter
D) recorder
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.5

6) One of the more formidable obstacles to mammalian gene cloning is the presence of
A) introns.
B) exons.
C) repressors.
D) integrators.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.3

7) What is the most important advantage of *Pfu* polymerase over *Taq* polymerase?
A) Unlike *Taq* polymerase, *Pfu* polymerase functions well at relatively high temperatures.
B) It is from a bacterium, not an archaean, so it is more effective for replicating eukaryotic DNA.
C) *Pfu* polymerase removes the need for primers during PCR.
D) Unlike *Taq* polymerase, *Pfu* polymerase has proofreading activity.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.1

8) What is the difference between PCR and RT-PCR?
A) Only PCR makes many copies of DNA rapidly.
B) RT-PCR uses an RNA template whereas PCR uses a DNA template.
C) Only PCR produces cDNA.
D) PCR uses a single stranded template whereas RT-PCR uses a double-stranded template.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.1

9) The genes encoding green fluorescent protein (GFP) and β-galactosidase are typically used in cloning as
A) transcription regulators.
B) global control genes.
C) promoter sequences.
D) reporter genes.
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.4

10) To verify a gene was cloned into a vector successfully, sequencing the vector as well as ________ are commonly performed.
A) agarose gel electrophoresis
B) fluorescence *in situ* hybridization
C) protein purification
D) northern blots
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.1

11) Which objective would be best to use a Southern blot rather than a Northern blot?
A) Determine if a gene is present in a genome.
B) Discover gene function.
C) Identify regulatory gene-protein interactions.
D) Quantify expression profiles of a gene.
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.1

12) A polymerase chain reaction (PCR) copies an individual gene segment *in vitro* with a(n) ________ primer(s).
A) individual RNA
B) individual DNA
C) pair of RNA
D) pair of DNA
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.1

13) Which of the following sequences is a palindrome, characteristic of many recognition sequences for restriction endonucleases?
A) TTGCCGA
AACGGCT
B) GGGGGGG
CCCCCCCC
C) GTAATG
CATTAC
D) GAATTC
CTTAAG
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.2

14) At which time period(s) during PCR thermocycling is/are hottest in temperature?
A) during DNA denaturation
B) during primer annealing
C) during primer extension/elongation
D) Both the first and last cycles are hotter in temperature than all other cycles.
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.1

15) To estimate the total concentration of a beneficial bacterial species in yogurt, ________ would provide the quickest results.
A) fluorescence *in situ* hybridization
B) qPCR
C) RT-PCR
D) a Southern blot
Answer: B
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.1

16) Which of the following is NOT a common step in molecular cloning using plasmids?
A) Fragment DNA into small segments.
B) Hybridize DNA sequences with slightly mismatched oligonucleotides.
C) Ligate DNA into vectors.
D) Insert the vectors into a host.
Answer: B
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.2

17) What molecular mechanism/feature does site-directed mutagenesis exploit to introduce a mutation at a specific site?
A) flanking complementary bound nucleotides permit non-complementary base pairing
B) methylated nucleotides disrupt DNA polymerase′s proofreading
C) nucleotide substitution when one is depleted
D) transposase-induced base pair changes
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.4

18) Inserting a kanamycin resistance cassette into a catabolic operon to confirm the gene is essential in degradation of a particular compound would involve all of the following EXCEPT
A) a reporter gene.
B) ligation.
C) recombination.
D) transformation.
Answer: A
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.4

19) Which statement is TRUE?
A) YACs are more likely than BACs to undergo recombination and rearrangement.
B) BACs are more likely than YACs to undergo recombination and rearrangement.
C) YACs and BACs undergo recombination and rearrangement at about the same rate.
D) It is impossible to state with any certainty whether YACs or BACs are more likely to undergo recombination and rearrangement, because environmental factors play a major role in the probability of one or the other occurring.
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.2

20) Which of those listed below is LEAST similar in what is being studied and concluded?
A) fluorescence *in situ* hybridization
B) GFP fusion protein
C) Northern blot
D) RT-PCR
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.1

21) The principle behind nucleic acid probe design is that the probe itself must contain
A) a key complementary part of the target gene sequence of interest.
B) all of the nucleotide sequence of the gene of interest to conclusively identify the gene.
C) an antibody to specifically bind to the gene of interest.
D) at least three separate complementary regions of the gene of interest.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.1

22) Which of those below is NOT an important consideration when designing a fusion protein construct?
A) Avoid hybridization of the fusion gene in the artificial construct.
B) Reading frame is the same for both the fusion gene and reporter gene.
C) Transcriptional start and stop signals are shared.
D) Translational start and stop signals are shared.
Answer: A
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 12.3

23) One challenge in cloning human somatotropin is that
A) it consists of multiple polypeptides, making synthesis complex.
B) its gene cannot be cloned as cDNA.
C) it can only be expressed by eukaryotic cells.
D) it is susceptible to digestion by bacterial proteases because it is a small protein hormone.
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.6

24) After digesting a DNA sequence, a restriction endonuclease can generate
A) blunt ends.
B) overhangs.
C) sticky ends.
D) blunt ends, overhangs, or sticky ends.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.2

25) What is the first step in constructing a metagenomic library from RNA?
A) The RNA must be converted to cDNA.
B) The RNA must be amplified through PCR, producing many RNA copies.
C) The RNA must be screened to identify the genes of interest.
D) The RNA must be inserted into plasmid vectors.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.9

26) The enzyme that covalently links both strands of a vector and inserted DNA in molecular cloning is
A) DNA ligase.
B) DNA phosphatase.
C) DNA hydrolase.
D) DNA transferase.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.2

27) Which of the following is NOT a limitation to using auxotrophs to prevent the spread of genetically modified genes to wild populations?
A) Auxotrophs may mutate in ways that allow them to synthesize the limiting nutrient.
B) Auxotrophs self-destruct using self-toxins when this method is applied.
C) Auxotrophs often cross-feed from the metabolites of other organisms in the environment.
D) Back mutation to the wild type may occur.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.13

28) The Ti plasmid is best suited for genetically manipulating
A) *Agrobacterium* spp.
B) fish.
C) plants.
D) viruses.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.7

29) Which of the following terms is used to describe a synthetic DNA fragment?
A) DNA cassette
B) DNA hybrid
C) recombinant DNA
D) artificial chromosome
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.4

30) Which process results in the production of a hybrid polypeptide?
A) vector fusion
B) operon fusion
C) protein fusion
D) translational fusion
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.5

31) If a protein that could be toxic to the expression host needs to be expressed in large quantities, then it is best to select an expression vector that
A) is not able to be replicated.
B) is inducible.
C) is attached to a normal cell promoter.
D) allows continual expression of the protein.
Answer: B
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 12.3

32) Some proteins expressed at high levels form inclusions that are relatively insoluble. What is the most effective way to facilitate purification of these proteins?
A) Use a reporter gene to locate the inclusion.
B) Decrease the number of insertions in the vector.
C) Produce the protein as a fusion protein.
D) Switch to an expression host with a larger intracellular volume.
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.3

33) The principle underlying how salmon were genetically engineered to grow faster is the
A) removal of a gene responsible for feeling full after eating.
B) replacement of inducible to constitutive hormone production.
C) resistance to bacterial infections which waste metabolic energy in the salmon to fight off.
D) addition of genes to enhance blood circulation and tissue development.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.7

34) Polyvalent vaccines using vaccinia virus are highly favored by doctors and physicians but are especially challenging for those who develop them, because
A) coat proteins form a relatively rigid structure and do not allow much space for additional proteins to be expressed.
B) multiple foreign proteins simultaneously synthesized often disrupts each other′s activity.
C) vaccinia and most other viruses engineered for vaccines contain only one restriction site for cloning in their genome.
D) virus genetic manipulation uses transfection, which is an inherently inefficient process.
Answer: B
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 12.8

35) Recognizing pathogens that contain multiple unique proteins which enable the human immune system to recognize just one and mount an effective response has opened the door on development of some vaccines only being
A) attenuated carrier viruses.
B) monovalent.
C) subunit vaccines.
D) purified protein administered.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.8

36) A poorly immunogenic vaccine often suggests the foreign proteins were not properly recognized by the immune system due to a lack of ________ necessary, which can also be engineered to occur with additional molecular manipulations.
A) complex folding
B) methylation
C) glucosylation
D) glycosylation
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.8

37) Which of the following is NOT an example of synthetic biology?
A) assembling gene sequences together into genome and creating a living organism from it
B) creating a new metabolic pathway that produces a previously unidentified compound
C) developing a novel polyvalent vaccine
D) making *Escherichia coli* photographic
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.11

38) Cloning vectors can be distinguished from expression vectors by
A) carrying *ori* genes for replication of the cloned sequence.
B) having a multiple cloning site (MCS).
C) having a high copy number per cell.
D) lacking a promoter site upstream of the insertion site.
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.2

39) Using a host defective in proteases is likely to be necessary when
A) engineering a complete metabolic pathway requiring several different enzymes.
B) overproducing proteins.
C) producing a small protein.
D) engineering transgenic animals with immune systems.
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.3

40) It is possible to rapidly screen for mutations in regulatory genes using
A) gene fusions.
B) defective proteases.
C) microinjection.
D) Southern blotting.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.5

41) What makes eukaryotic transcripts easier to isolate than bacterial transcripts?
A) Eukaryotic transcripts are not methylated but their genes are often methylated.
B) Larger transcript size in eukaryotes enables easy size-selection methods.
C) mRNA is polyadenylated in eukaryotes.
D) Transcripts are the most abundant RNAs in eukaryotes.
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.3

12.2 True/False Questions

1) A common reporter protein is green fluorescent protein (GFP).
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.5

2) High expression levels of a eukaryotic gene in a bacterium such as *Escherichia coli* cannot be accomplished due to the presence of introns.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.3

3) The key steps in cloning a foreign gene into a vector, regardless of the application, involve isolating the insert fragment, ligating the insert into a vector, and transforming it into a host.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.3

4) Strong promoters used for genetic manipulation are usually regulated by specific molecules.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.3

5) DNA polymerases from *Escherichia coli* cannot be used to artificially copy gene sequences with a thermocycler.
Answer: TRUE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.1

6) Although various codons often code for the same amino acid, it is important to choose the codon preferred by the expression host itself.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.3

7) The *lacZ* gene is commonly used as a reporter gene, because its substrate lactose is well known and easily measured.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.5

8) One problem with both BACs and YACs is that genetic regions of these chromosomes cannot be subcloned.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.2

9) An operon fusion has a transcriptional signal from one gene fused with the translational start and coding sequence from another.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.5

10) One fundamental technique of genetic engineering includes the ability to cut DNA into random fragments.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.2

11) Modification enzymes typically methylate specific bases within the recognition sequence to prevent digestion of the nucleotide sequence by restriction endonucleases.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.2

12) Engineering a metabolic pathway enables a researcher to use different genes from unrelated organisms.
Answer: FALSE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.11

13) Developing vaccines for humans relies heavily on manipulating and engineering vectors.
Answer: TRUE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.8

14) One important advantage of eukaryotic cells as hosts for cloning vectors is that they already possess the complex RNA and posttranslational processing systems required for the production of eukaryotic proteins.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.2

15) Due to well-developed molecular tools and careful screening designs, functional genes can be isolated directly by isolation from the environment rather than cultivating the diverse species in a microbial community.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.9

16) Regardless of the DNA polymerase used in PCR, such as *Taq* or *Pfu*, they all have an inherent inability to perfectly copy the template strand, which means the polymerases themselves occasionally make mutations in the sequences they copy.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.1

17) DNA ligase mediates the insertion of foreign DNA into a vector, but it will only be able to do so if the inserts and vector have matching sticky or blunt ends.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.2

18) An effective way to introduce DNA into plant cells is using the Ti plasmid, which comes from a plant pathogen.
Answer: TRUE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.7

19) Genetically modified plants resistant to the herbicide glyphosphate contain a gene from Bacillus thuringiensis.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.7

20) Green fluorescent protein (GFP) is used for detecting translational activity of a fused protein, whereas *lacZ* reporters are used to detect transcriptional activity of a fused gene.
Answer: FALSE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.5

21) One method to circumvent issues with introns when expressing a eukaryotic gene in a bacterium is to simply clone the mature transcript.
Answer: TRUE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.3

22) If vaccinia viruses would not both be immunogenic and relatively benign, they would likely not be a favored vehicle for vaccinations.
Answer: TRUE
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 12.8

23) PCR rapidly produces many copies of entire DNA molecules.
Answer: FALSE
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 12.1

12.3 Essay Questions

1) Create an experiment where you would use both a Southern blot and a Northern blot to determine if a newly isolated bacterium contains and actively uses the same catabolic pathway as previously demonstrated. What is the value in doing the Southern blot first? With results from these experiments, how would you conclude if the same pathway is active or not?
Answer: Probes would be designed off gene sequences from the previously demonstrated pathway, and the same DNA probes could be used for both the Southern and Northern blots. With isolated genomic DNA from the isolated bacterium, the probes would be applied to visualize the presence of each gene. All probes hybridizing to the genome suggests the pathway exists in the isolate, and the probes would then be used to determine is those genes were all active during the catabolism of the particular substrate. The substrate inducing the catabolic pathway would be provided to the isolate as a sole carbon source and RNA harvested to determine which transcripts are present. Activity of the genes for the pathway would be identified with probes again hybridizing to the same sequences as in the Southern. The Northern would not have been necessary to perform if all the genes were not first identified in the genome. To conclude the same genes are responsible for the catabolic pathway as previously documented, all gene/transcript probes would have to hybridize to the transcriptome.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 12.1

2) Describe the steps of gene cloning using a vector.
Answer: DNA is amplified. It then must be inserted into a vector and then the vector must be inserted into a host cell.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.2

3) Explain how eukaryotic genes can be used to produce proteins in a bacterial host. What steps are required?
Answer: Because eukaryotic RNA is processed to form mRNA, it is important to use eukaryotic mRNA to make cDNA before the genes are inserted into a bacterial host. The eukaryotic mRNA can be extracted using chromatography followed by the use of a low salt buffer to remove the mRNA from the chromatography column. RT-PCR is used to make cDNA, which is inserted into an expression vector.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.3

4) Describe three characteristics of an ideal cloning host for genetic engineering.
Answer: Answers will vary. Characteristics of a host that is easy to manipulate and work with include the ability to efficiently take up DNA, to contain the necessary genes to encode for enzymes that facilitate replication of the vector, to grow in an inexpensive medium, nonpathogenic, rapid growth, and stability in culture situations.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.2

5) Explain how T7 promoters regulate transcription and why cloning vectors contain them.
Answer: The bacteriophage T7 polymerase gene is typically cloned on a vector such that it is under regulation of the *lac* promoter. In this way, analogs of lactose such as IPTG can be exogenously added to the growth medium at any given time to induce expression of the T7 polymerase. The T7 polymerase will recognize only T7 promoters, so only genes downstream to them are turned on in response to IPTG. Many cloning vectors contain T7 promoters upstream of the insert site so that their transcription can be easily manipulated.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.3

6) Defend why codon usage must be considered when cloning a gene into another organism.
Answer: The primary concern for codon usage is that one organism might natively make a large quantity of a particular amino acid, so proteins rich in this particular amino acid can maintain a high level of translation, yet the cloning host does not make high levels of the amino acid. This results in unexpectedly low levels of the engineered gene product.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 12.3

7) When trying to make a mammalian protein in a bacterium, expression vectors are often used. Using your knowledge of prokaryotic and eukaryotic genetics, explain why expression vectors are so important.
Answer: One issue with transcription of the cloned gene(s) is the differences in RNA polymerases and their recognition (promoter) sequences. For example, strong eukaryotic promoters might not function inside a bacterial cell. Another problem is that origins of replication in vectors are not universal to hosts, meaning that plasmids that replicate in eukaryotes cannot necessarily replicate in *E. coli*. Most importantly, a fundamental difference in expressing eukaryotic genes in prokaryotes is that prokaryotes generally lack the machinery required to splice out introns.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 12.3

8) Explain how a metagenomic DNA library is created and what type of vector is needed.
Answer: Answers will vary, but a common method is to obtain DNA from the environment, digest the DNA sample, and clone the fragments into a vector. BACs are commonly used because they can carry relatively large DNA inserts.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.9

9) What are some advantages and limitations of using biofuels, and why are genetically engineered microbes more useful for their production than naturally occurring microbes?
Answer: Biofuels provide renewable sources of energy and provide additional fuel options to help address environmental concerns. However, biofuels are currently produced on a relatively small scale and considerable expense would be required to scale up production. The microorganisms that produce biofuels naturally may also produce toxic byproducts and sometimes lack enzymes for critical steps, so genetically modified microbes are needed to maximize production.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.11

10) What is necessary for a YAC to function as a normal eukaryotic chromosome?
Answer: A YAC must contain an origin of replication (*ori)* specific to the host, telomeres to ensure the DNA termini are replicated, and centromeres for segregation during mitosis.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.2

11) Describe a method developed by molecular biologists to easily observe the success of a genetic engineering procedure involving the ligating of a gene of interest into a plasmid.
Answer: Answers will vary, but two possible answers are reporter constructs where reporter proteins are observed (e.g., fluorescence) and blue-white screening of the colonies themselves for disruption/loss of the LacZ reporter.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.5

12) Scientists produced a synthetic bacterium in 2010, but no one has yet produced an entire cell from scratch. How did the synthetic bacterium from 2010 compare with an entirely de novo cell, and what could be reasons that scientists might want to accomplish the latter?
Answer: Answers will vary. The synthetic bacterium was constructed by producing a synthetic chromosome that was inserted into a bacterial cell. The cell already contained structures such as ribosomes and enzymes. An entirely synthetic cell would be entirely assembled from "biobricks," meaning that all parts would be synthetic (not just the chromosome). Students can speculate about reasons that such a synthetic organism could be useful.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 12.11

13) Describe the three main steps to clone a gene into an organism.
Answer: First, the DNA sequence of interest (gene) must be isolated. The isolated DNA is often larger than the gene of interest, so cutting the short gene sequence out of the large sequence is often performed, which uses restriction enzymes. Inserting the gene into a vector is the second main step. This usually requires DNA ligase whether the insert and vector are blunt- or sticky-ended. The final step involves transforming an (host) organism to house the clone vector.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.2

14) Explain why DNA fragments migrate toward the positive electrode and why some fragments migrate more rapidly than others during gel electrophoresis. How does this electronegativity influence observations and conclusions drawn from DNA migrated through an agarose gel?
Answer: DNA is negatively charged due to its phosphate backbone, and because opposite charges are attracted to each other DNA migrates towards the positive electrode. A large DNA size (or length) runs slower on a gel despite having a stronger charge than a smaller fragment, because the agarose (or polyacrylamide) within the gel acts as a physical barrier to impede and slow down the larger DNA molecules.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 21.1

15) When is it appropriate to use an artificial chromosome vector? Describe a specific example.
Answer: Artificial chromosome vectors are generally best suited for carrying large inserts of DNA compared to other traditional plasmids. During DNA sequencing of an organism, several overlapping reads are necessary to develop a consensus sequence and therefore mandates a large number of different inserts to be sequenced. With the example of the human genome, an immense number of 1,000 bp fragments would need to be sequenced, all housed within an isolated bacterium containing the plasmid. Larger artificial chromosome decrease the complexity of genome sequencing projects by containing larger inserts (e.g., 800,000 bp fragments) so that fewer individual cells and fewer artificial chromosome vectors need to be sequenced.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.2

16) Compare how the process of cloning differs when a vector with sticky ends is used and when a vector with blunt ends is used.
Answer: All blunt-ended products are capable of joining together, however not all sticky ends will bind to each other. For example, an overhang of one G requires that another product must contain one nucleotide overhang and must be the complementary nucleotide (C). This usually means sticky-ended products must be digested by the same restriction endonuclease.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.2

17) Describe the usefulness of blue-white screening, also called α-complementation, in cloning vectors such as pUC19. Include in your answer the terms polylinker, DNA ligase, *lacZ* gene, insertional inactivation, Xgal, and β-galactosidase.
Answer: This is a powerful screening method, because white colonies contain a gene inserted into their plasmid whereas blue colonies do not. In the example of pUC19, the *lacZ* gene contains a polylinker site where the plasmid can be digested and have a foreign gene cloned into the site. The plasmid is re-circularized by DNA ligase such that now the *lacZ* gene and therefore β-galactosidase (LacZ) activity is disrupted. When transformed cells are grown on a medium with the artificial Xgal substrate, all of those without an insert will have a functional LacZ and will convert the Xgal to a blue color. A white colony suggests the cells contain an inserted gene within the *lacZ* gene of the plasmid, and these white colonies are likely to be the ones of interest.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 12.2

18) Explain the process of site-directed mutagenesis, and discuss some applications of this technique.
Answer: Site-directed mutagenesis involves changing individual nucleotides at precise locations within a sequence *in vitro*. Using short synthetic oligonucleotides (primers) with a mismatched base pair, the mutation site will be amplified with several rounds of PCR. An application to this method is to change nucleotides so that certain amino acids are changed. This circumvents issues associated with codon bias when cloning a gene into a host with a different codon usage pattern. Changing specific nucleotides could also change the recognition site for a restriction endonuclease to either create or remove a specific restriction site. A third application is to introduce several point mutations for the disruption of a gene to study its loss in phenotype and therefore identify its function.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.4

19) Compare and contrast operon and protein fusions.
Answer: Operon and protein fusions are both examples of gene fusions that bring together two separate nucleotide sequences. A gene fusion is formed when one of the genes′ promoter regions is fused to the coding region of the other gene upstream of it. Protein fusions have transcription occur at the same site for the two fused genes, and translation of the fused transcript takes place at one site. An operon fusion retains separate translational start sites despite sharing the same transcriptional start site.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 12.5